MATH 102 Calculus II

Final Exam (mock) – Solutions

Problem 1

Let
$$D = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 \le 4,\ y^2 + z^2 \le 4\}.$$
Then D is the intersection of the two cylinders $x^2 + y^2 \le 4$ and $y^2 + z^2 \le 4$. For each fixed $y \in [-2, 2]$ we have

$$x^2 \le 4 - y^2, \qquad z^2 \le 4 - y^2.$$

Thus
$$-\sqrt{4-y^2} \le x \le \sqrt{4-y^2}, \qquad -\sqrt{4-y^2} \le z \le \sqrt{4-y^2}.$$
Hence the volume is

$$\begin{aligned}
\mathrm{Vol}(D) &= \int_{-2}^{2}\int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}}\int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} 1\,\mathrm{d}z\,\mathrm{d}x\,\mathrm{d}y \\
&= \int_{-2}^{2}\int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} 2\sqrt{4-y^2}\,\mathrm{d}x\,\mathrm{d}y \\
&= \int_{-2}^{2} 2\sqrt{4-y^2}\cdot\left(2\sqrt{4-y^2}\right)\mathrm{d}y \\
&= \int_{-2}^{2} 4(4-y^2)\,\mathrm{d}y = 4\left[4y - \frac{y^3}{3}\right]_{-2}^{2} = 4\left(\frac{32}{3}\right) = \frac{128}{3}.
\end{aligned}$$

$$\boxed{\mathrm{Vol}(D) = \frac{128}{3}}.$$

Problem 2

Let
$$D = \{(x, y) \in \mathbb{R}^2 : 9(x-1)^2 + 4(y-3)^2 \le 36\}.$$
This is the ellipse centered at $(1, 3)$ with semi-axes $a = 2$ (in the x-direction) and $b = 3$ (in the y-direction).
Translate the coordinates:
$$u = x - 1, \qquad v = y - 3.$$
Then D becomes the ellipse
$$\frac{u^2}{2^2} + \frac{v^2}{3^2} \le 1.$$
Moreover
$$x + y = (u + 1) + (v + 3) = u + v + 4.$$
Hence
$$\int_D (x + y)\,\mathrm{d}A = \int_{D'} (u + v + 4)\,\mathrm{d}A,$$
where D' is the same ellipse written in (u, v)-coordinates.
Because D' is centrally symmetric about $(0, 0)$, the integrals of the odd functions u and v over D' vanish:
$$\int_{D'} u\,\mathrm{d}A = 0, \qquad \int_{D'} v\,\mathrm{d}A = 0.$$
Therefore
$$\int_D (x + y)\,\mathrm{d}A = 4\ \mathrm{Area}(D).$$
The area of an ellipse with semi-axes a and b is πab, so
$$\mathrm{Area}(D) = \pi \cdot 2 \cdot 3 = 6\pi,$$
and hence
$$\int_D (x + y)\,\mathrm{d}A = 4 \cdot 6\pi = 24\pi.$$
$$\boxed{\int_D (x + y)\,\mathrm{d}A = 24\pi.}$$

Problem 3

The solid E is the intersection of the ball of radius R with the cone of opening angle ϕ_0 about the positive z-axis. In spherical coordinates (ρ, ϕ, θ) (with $\rho \geq 0$, $0 \leq \phi \leq \pi$ measured from the positive z-axis, and $0 \leq \theta < 2\pi$),

$$E = \{0 \leq \rho \leq R,\ 0 \leq \phi \leq \phi_0,\ 0 \leq \theta < 2\pi\}.$$

The distance from (x, y, z) to the origin is simply ρ. The average distance is

$$\overline{d} = \frac{1}{\mathrm{Vol}(E)} \int_E \rho \,\mathrm{d}V.$$

First compute the volume:

$$\begin{aligned}
\mathrm{Vol}(E) &= \int_0^{2\pi} \int_0^{\phi_0} \int_0^R \rho^2 \sin\phi \,\mathrm{d}\rho \,\mathrm{d}\phi \,\mathrm{d}\theta \\
&= \left(\int_0^{2\pi} \mathrm{d}\theta\right) \left(\int_0^{\phi_0} \sin\phi \,\mathrm{d}\phi\right) \left(\int_0^R \rho^2 \,\mathrm{d}\rho\right) \\
&= 2\pi\,(1 - \cos\phi_0)\,\frac{R^3}{3} = \frac{2\pi R^3}{3}(1 - \cos\phi_0).
\end{aligned}$$

Next compute the numerator:

$$\begin{aligned}
\int_E \rho \,\mathrm{d}V &= \int_0^{2\pi} \int_0^{\phi_0} \int_0^R \rho \cdot \rho^2 \sin\phi \,\mathrm{d}\rho \,\mathrm{d}\phi \,\mathrm{d}\theta \\
&= \left(\int_0^{2\pi} \mathrm{d}\theta\right) \left(\int_0^{\phi_0} \sin\phi \,\mathrm{d}\phi\right) \left(\int_0^R \rho^3 \,\mathrm{d}\rho\right) \\
&= 2\pi\,(1 - \cos\phi_0)\,\frac{R^4}{4} = \frac{\pi R^4}{2}(1 - \cos\phi_0).
\end{aligned}$$

Thus

$$\overline{d} = \frac{\frac{\pi R^4}{2}(1 - \cos\phi_0)}{\frac{2\pi R^3}{3}(1 - \cos\phi_0)} = \frac{3}{4}R.$$

$$\boxed{\text{The average distance } \overline{d} = \frac{3R}{4}.}$$

Problem 4

Let C be the closed curve in the xy-plane given by

$$y^2 = x^3 + x^2, \qquad -1 \le x \le 0,$$

with both upper and lower branches, oriented counterclockwise. Let D be the region enclosed by C.
We are given the force field

$$\mathbf{F} = (P, Q) = (2xy, -x^3).$$

The work done is

$$W = \int_C \mathbf{F} \cdot T \mathrm{d}s = \int_C P \,\mathrm{d}x + Q \,\mathrm{d}y.$$

By Green's theorem (with positive, i.e. counterclockwise orientation),

$$W = \int_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathrm{d}A.$$

We compute

$$\frac{\partial Q}{\partial x} = -3x^2, \qquad \frac{\partial P}{\partial y} = 2x,$$

so

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = -3x^2 - 2x = -(3x^2 + 2x).$$

The region D can be described as

$$D = \{(x, y) \mid -1 \le x \le 0, \ -\sqrt{x^3 + x^2} \le y \le \sqrt{x^3 + x^2}\}.$$

Therefore

$$\begin{aligned} W &= \int_{-1}^{0} \int_{-\sqrt{x^3+x^2}}^{\sqrt{x^3+x^2}} -(3x^2 + 2x) \,\mathrm{d}y \,\mathrm{d}x \\ &= \int_{-1}^{0} -(3x^2 + 2x) \cdot 2\sqrt{x^3 + x^2} \,\mathrm{d}x \\ &= \int_{-1}^{0} -2(3x^2 + 2x)\sqrt{x^3 + x^2} \,\mathrm{d}x. \end{aligned}$$

Let $u = x^3 + x^2$ so that $\mathrm{d}u = (3x^2 + 2x) \,\mathrm{d}x$. Then

$$W = \int_{-1}^{0} -2\sqrt{x^3 + x^2} \,(3x^2 + 2x) \,\mathrm{d}x = \int -2\sqrt{u} \,\mathrm{d}u = -\frac{4}{3}u^{3/2} + C.$$

Thus an antiderivative is

$$F(x) = -\frac{4}{3}(x^3 + x^2)^{3/2},$$

and

$$W = F(0) - F(-1) = -\frac{4}{3} \cdot 0^{3/2} - \left(-\frac{4}{3} \cdot 0^{3/2} \right) = 0.$$

$$\boxed{W = 0.}$$

Problem 5

We have
$$\mathbf{F}(x,y) = \big(ye^{xy},\, xe^{xy}\big).$$
Set $P(x,y) = ye^{xy}$, $Q(x,y) = xe^{xy}$.
Compute the mixed partial derivatives:
$$\frac{\partial P}{\partial y} = e^{xy} + xye^{xy}, \qquad \frac{\partial Q}{\partial x} = e^{xy} + xye^{xy}.$$
Thus $\partial P/\partial y = \partial Q/\partial x$ on all of $\mathbb{R}^2$.
Since the domain $\mathbb{R}^2$ is simply connected, this implies that $\mathbf{F}$ is conservative. Therefore, for a closed curve, the integral is 0.
The rectangle in the problem is a closed curve C, hence
$$\boxed{\int_C \mathbf{F}\cdot T\mathrm{d}s = 0.}$$

Problem 6

Let $\mathbf{F}(x, y, z) = (x, y, z)$ and

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 9,\ z \geq 0\}$$

be the upper hemisphere of radius 3.
We take S with the outward (radial) orientation. For a pint (x, y, z) on S, the outward unit normal is

$$\mathbf{n} = \frac{1}{3}(x, y, z),$$

since the radius is 3. Then

$$\mathbf{F} \cdot \mathbf{n} = (x, y, z) \cdot \frac{1}{3}(x, y, z) = \frac{x^2 + y^2 + z^2}{3} = \frac{9}{3} = 3.$$

Thus $\mathbf{F} \cdot \mathbf{n}$ is constant on S.
The area of the hemisphere of radius 3 is

$$\text{Area}(S) = 2\pi R^2 = 2\pi \cdot 3^2 = 18\pi.$$

Hence the flux through S is

$$\int_S \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}S = \int_S 3 \, \mathrm{d}S = 3 \cdot \text{Area}(S) = 3 \cdot 18\pi = 54\pi.$$

$$\boxed{\int_S \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}S = 54\pi.}$$

Problem 7

We are given

$$\mathbf{F}(x, y, z) = \frac{(x, y, z)}{(x^2 + y^2 + z^2)^{3/2}}, \qquad (x, y, z) \neq (0, 0, 0).$$

Suppose, for contradiction, that there exists a vector field $\mathbf{G}$ on $\mathbb{R}^3 \setminus \{0\}$ such that

$$\mathbf{F} = \nabla \times \mathbf{G}.$$

Step 1: A property of curls

Let S be a smooth closed surface lying in an open region where $\mathbf{G}$ is defined and smooth. We claim that

$$\int_S (\nabla \times \mathbf{G}) \cdot \mathbf{n} \, \mathrm{d}S = 0.$$

To see this for a sphere, let S be a sphere and split it into the upper hemisphere S_1 and the lower hemisphere S_2, whose common boundary is the equator C.
By Stokes' theorem,

$$\int_{S_1} (\nabla \times \mathbf{G}) \cdot \mathbf{n} \, \mathrm{d}S = \int_C \mathbf{G} \cdot T \mathrm{d}s,$$

where C is oriented as the boundary of S_1. For S_2 (oriented outward), the induced orientation on C is the opposite, hence

$$\int_{S_2} (\nabla \times \mathbf{G}) \cdot \mathbf{n} \, \mathrm{d}S = -\int_C \mathbf{G} \cdot T \mathrm{d}s.$$

Adding these gives

$$\int_S (\nabla \times \mathbf{G}) \cdot \mathbf{n} \, \mathrm{d}S = \int_{S_1} \cdots + \int_{S_2} \cdots = \int_C \mathbf{G} \cdot T \mathrm{d}s - \int_C \mathbf{G} \cdot T \mathrm{d}s = 0.$$

Thus, for any such S,

$$\int_S (\nabla \times \mathbf{G}) \cdot \mathbf{n} \, \mathrm{d}S = 0.$$

Step 2: Flux of F through a sphere

Take S_R to be a sphere of radius $R > 0$ centered at the origin:

$$S_R = \{(x, y, z) \mid x^2 + y^2 + z^2 = R^2\}.$$

Note that $S_R \subset \mathbb{R}^3 \setminus \{0\}$, so both $\mathbf{F}$ and (by assumption) $\mathbf{G}$ are defined and smooth on and near S_R.
On S_R we have $r = \sqrt{x^2 + y^2 + z^2} = R$, and the outward unit normal is

$$\mathbf{n} = \frac{(x, y, z)}{R} = \frac{\mathbf{r}}{R}.$$

Also,

$$\mathbf{F} = \frac{\mathbf{r}}{R^3}.$$

Therefore

$$\mathbf{F} \cdot \mathbf{n} = \frac{\mathbf{r}}{R^3} \cdot \frac{\mathbf{r}}{R} = \frac{|\mathbf{r}|^2}{R^4} = \frac{R^2}{R^4} = \frac{1}{R^2}.$$

This is constant on S_R, so

$$\int_{S_R} \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}S = \frac{1}{R^2} \operatorname{Area}(S_R) = \frac{1}{R^2} \cdot 4\pi R^2 = 4\pi.$$

Step 3: Contradiction

If $\mathbf{F} = \nabla \times \mathbf{G}$, then by Step 1 applied to S_R,

$$\int_{S_R} \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}S = \int_{S_R} (\nabla \times \mathbf{G}) \cdot \mathbf{n} \, \mathrm{d}S = 0.$$

But Step 2 shows that this flux equals $4\pi \neq 0$. This is a contradiction. Therefore no such vector field $\mathbf{G}$ can exist.

$$\boxed{\text{There is no } \mathbf{G} \text{ on } \mathbb{R}^3 \setminus \{0\} \text{ with } \mathbf{F} = \nabla \times \mathbf{G}.}$$

Problem 8

Let S be the part of the paraboloid $z = 1 - x^2 - y^2$ above the xy-plane (so $z \ge 0$), and $\mathbf{F} = (x, y, 2(1-z))$.
The projection of S onto the xy-plane is the disk

$$D = \{(x, y) \mid x^2 + y^2 \le 1\}.$$

On S we have $z = 1 - x^2 - y^2$.

(a) Direct computation

Write $z = 1 - x^2 - y^2$. Then

$$\frac{\partial z}{\partial x} = -2x, \qquad \frac{\partial z}{\partial y} = -2y.$$

For the upward orientation, the vector $\mathbf{N}\,\mathrm{d}\sigma$ is

$$\mathbf{N}\,\mathrm{d}\sigma = (-z_x, -z_y, 1)\,\mathrm{d}x\,\mathrm{d}y = (2x, 2y, 1)\,\mathrm{d}x\,\mathrm{d}y.$$

On S,

$$\mathbf{F}(x, y, z) = \big(x,\ y,\ 2(1-z)\big) = \big(x,\ y,\ 2(x^2+y^2)\big),$$

since $1 - z = x^2 + y^2$ on S.
Therefore

$$\begin{aligned}
\mathbf{F}\cdot\mathbf{N}\,\mathrm{d}\sigma &= (x, y, 2(x^2+y^2))\cdot(2x, 2y, 1)\,\mathrm{d}x\,\mathrm{d}y \\
&= \big(2x^2 + 2y^2 + 2(x^2+y^2)\big)\,\mathrm{d}x\,\mathrm{d}y \\
&= 4(x^2+y^2)\,\mathrm{d}x\,\mathrm{d}y.
\end{aligned}$$

Thus the flux is

$$\int_S \mathbf{F}\cdot\mathbf{N}\,\mathrm{d}\sigma = \int_D 4(x^2+y^2)\,\mathrm{d}x\,\mathrm{d}y.$$

Use polar coordinates on D:

$$x = r\cos\theta, \quad y = r\sin\theta, \quad 0 \le r \le 1,\ 0 \le \theta < 2\pi,$$

with $\mathrm{d}x\,\mathrm{d}y = r\,\mathrm{d}r\,\mathrm{d}\theta$. Then $x^2 + y^2 = r^2$, so

$$\begin{aligned}
\int_D 4(x^2+y^2)\,\mathrm{d}x\,\mathrm{d}y &= \int_0^{2\pi}\int_0^1 4r^2\cdot r\,\mathrm{d}r\,\mathrm{d}\theta \\
&= \int_0^{2\pi}\int_0^1 4r^3\,\mathrm{d}r\,\mathrm{d}\theta \\
&= \int_0^{2\pi}\big[r^4\big]_0^1\,\mathrm{d}\theta = \int_0^{2\pi} 1\,\mathrm{d}\theta = 2\pi.
\end{aligned}$$

Hence

$$\boxed{\int_S \mathbf{F}\cdot\mathbf{N}\,\mathrm{d}\sigma = 2\pi.}$$

(b) Using the divergence theorem

Let V be the solid bounded above by S and below by the disk

$$B = \{(x, y, 0) \mid x^2 + y^2 \leq 1\}$$

in the plane $z = 0$. We orient the boundary $\partial V = S \cup B$ outward (i.e. away from V). Compute the divergence of $\mathbf{F}$:

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}x + \frac{\partial}{\partial y}y + \frac{\partial}{\partial z}\big(2(1-z)\big) = 1 + 1 - 2 = 0.$$

By the divergence theorem,

$$\int_{\partial V} \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}S = \int_V \nabla \cdot \mathbf{F} \, \mathrm{d}V = 0.$$

Thus

$$\int_S \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}S + \int_B \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}S = 0.$$

On B the outward normal (with respect to V, which lies above the xy-plane) is $\mathbf{n}_B = (0, 0, -1)$, and $z = 0$. Hence

$$\mathbf{F}(x, y, 0) = (x, y, 2), \qquad \mathbf{F} \cdot \mathbf{n}_B = (x, y, 2) \cdot (0, 0, -1) = -2.$$

So

$$\int_B \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}S = \int_{x^2+y^2 \leq 1} -2 \, \mathrm{d}x \, \mathrm{d}y = -2 \cdot \pi \cdot 1^2 = -2\pi.$$

Therefore

$$\int_S \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}S = -\int_B \mathbf{F} \cdot \mathbf{n} \, \mathrm{d}S = -(-2\pi) = 2\pi,$$

in agreement with part (a).

Problem 9

The vector field

$$\mathbf{F}(x,y) = \left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2}\right), \qquad (x,y) \neq (0,0),$$

is defined on $\mathbb{R}^2 \setminus \{(0,0)\}$.
First compute the scalar curl (in 2D):

$$P(x,y) = \frac{-y}{x^2+y^2}, \qquad Q(x,y) = \frac{x}{x^2+y^2}.$$

Then

$$\frac{\partial P}{\partial y} = \frac{-1\cdot(x^2+y^2) - (-y)\cdot 2y}{(x^2+y^2)^2} = \frac{-x^2+y^2}{(x^2+y^2)^2},$$

and

$$\frac{\partial Q}{\partial x} = \frac{1\cdot(x^2+y^2) - x\cdot 2x}{(x^2+y^2)^2} = \frac{-x^2+y^2}{(x^2+y^2)^2}.$$

Hence

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0$$

on the domain. So locally $\mathbf{F}$ looks like a gradient field wherever the domain is simply connected.
However, $\mathbf{F}$ is *not* conservative on its full domain $\mathbb{R}^2 \setminus \{0\}$, because the domain is not simply connected and there exists a closed loop with nonzero integral.
Consider the unit circle C around the origin:

$$\mathbf{X}(t) = (\cos t, \sin t), \quad 0 \leq t \leq 2\pi,$$

oriented counterclockwise. Then

$$T\mathrm{d}s = (-\sin t, \cos t)\,\mathrm{d}t,$$

and

$$\mathbf{F}(\cos t, \sin t) = \left(\frac{-\sin t}{\cos^2 t + \sin^2 t}, \frac{\cos t}{\cos^2 t + \sin^2 t}\right) = (-\sin t, \cos t).$$

Therefore

$$\mathbf{F}\cdot T\mathrm{d}s = (-\sin t, \cos t)\cdot(-\sin t, \cos t)\,\mathrm{d}t = (\sin^2 t + \cos^2 t)\,\mathrm{d}t = \mathrm{d}t.$$

So

$$\int_C \mathbf{F}\cdot T\mathrm{d}s = \int_0^{2\pi} \mathrm{d}t = 2\pi \neq 0.$$

If $\mathbf{F}$ were conservative, the integral around any closed curve would be 0. Hence $\mathbf{F}$ is not conservative on its domain.

$$\boxed{\mathbf{F} \text{ is not conservative on } \mathbb{R}^2 \setminus \{0\}.}$$

Problem 10

Let S be a part of the surface of the cone

$$x = 1 - \sqrt{y^2 + z^2}, \quad 0 \le x \le 1, \quad z \ge 0,$$

and let the vector field $\mathbf{F} = (xz,\ xy^2 + 2z,\ xy + z)$. Suppose that S is oriented by a unit normal field $\mathbf{N}$ such that $\mathbf{N} \cdot e_3 > 0$.

(a) Compute $\displaystyle\int_S (\mathrm{curl}\mathbf{F}) \cdot \mathbf{N}\, \mathrm{d}\sigma$

First compute the curl:

$$\mathrm{curl}\mathbf{F} = \nabla \times \mathbf{F} = \Big(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\Big).$$

We have

$$F_1 = xz, \quad F_2 = xy^2 + 2z, \quad F_3 = xy + z,$$

so

$$\begin{aligned} \frac{\partial F_3}{\partial y} &= x, & \frac{\partial F_2}{\partial z} &= 2, \\ \frac{\partial F_1}{\partial z} &= x, & \frac{\partial F_3}{\partial x} &= y, \\ \frac{\partial F_2}{\partial x} &= y^2, & \frac{\partial F_1}{\partial y} &= 0. \end{aligned}$$

Thus

$$\mathrm{curl}\mathbf{F}(x, y, z) = (x - 2,\ x - y,\ y^2).$$

Now parametrize S. Let

$$r = \sqrt{y^2 + z^2}, \quad 0 \le r \le 1, \quad 0 \le \theta \le \pi,$$

and set

$$x = 1 - r, \quad y = r\cos\theta, \quad z = r\sin\theta.$$

Then $z \ge 0$ and $y^2 + z^2 = r^2 = (1 - x)^2$, $0 \le x \le 1$.
Define

$$\mathbf{r}(r, \theta) = \big(1 - r,\ r\cos\theta,\ r\sin\theta\big).$$

Compute

$$\mathbf{r}_r = (-1, \cos\theta, \sin\theta), \quad \mathbf{r}_\theta = (0, -r\sin\theta, r\cos\theta).$$

Then

$$\begin{aligned} \mathbf{r}_r \times \mathbf{r}_\theta &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & \cos\theta & \sin\theta \\ 0 & -r\sin\theta & r\cos\theta \end{vmatrix} \\ &= r\,(1, \cos\theta, \sin\theta), \end{aligned}$$

whose z-component is $r\sin\theta \ge 0$ for $0 \le \theta \le \pi$, so this orientation agrees with $\mathbf{N} \cdot e_3 > 0$.
Thus

$$\mathbf{N}\, \mathrm{d}\sigma = \mathbf{r}_r \times \mathbf{r}_\theta\, \mathrm{d}r\, \mathrm{d}\theta = r\,(1, \cos\theta, \sin\theta)\, \mathrm{d}r\, \mathrm{d}\theta.$$

On S,

$$x = 1 - r, \quad y = r\cos\theta, \quad z = r\sin\theta,$$

so

$$\operatorname{curl}\mathbf{F}\big(\mathbf{r}(r,\theta)\big) = \big((1-r)-2,\ (1-r) - r\cos\theta,\ r^2\cos^2\theta\big) = \big(r-1,\ 1 - r(1+\cos\theta),\ r^2\cos^2\theta\big)\,.$$

Hence

$$\begin{aligned}
(\operatorname{curl}\mathbf{F})\cdot\mathbf{N}\,\mathrm{d}\sigma &= \operatorname{curl}\mathbf{F}\big(\mathbf{r}(r,\theta)\big)\cdot\big(\mathbf{r}_r \times \mathbf{r}_\theta\big)\,\mathrm{d}r\,\mathrm{d}\theta \\
&= r\Big[(r-1)\cdot 1 + \big(1 - r(1+\cos\theta)\big)\cos\theta + r^2\cos^2\theta\sin\theta\Big]\mathrm{d}r\,\mathrm{d}\theta \\
&= r\Big(r^2\sin\theta\cos^2\theta - r - \big(r\cos\theta + r - 1\big)\cos\theta - 1\Big)\mathrm{d}r\,\mathrm{d}\theta.
\end{aligned}$$

Expanding,

$$(\operatorname{curl}\mathbf{F})\cdot\mathbf{N}\,\mathrm{d}\sigma = \big(r^3\sin\theta\cos^2\theta - r^2\cos^2\theta - r^2\cos\theta - r^2 + r\cos\theta - r\big)\mathrm{d}r\,\mathrm{d}\theta.$$

Thus

$$\int_S (\operatorname{curl}\mathbf{F})\cdot\mathbf{N}\,\mathrm{d}\sigma = \int_0^\pi\int_0^1 \big(r^3\sin\theta\cos^2\theta - r^2\cos^2\theta - r^2\cos\theta - r^2 + r\cos\theta - r\big)\mathrm{d}r\,\mathrm{d}\theta.$$

First integrate in r:

$$\begin{aligned}
&\int_0^1 \big(r^3\sin\theta\cos^2\theta - r^2\cos^2\theta - r^2\cos\theta - r^2 + r\cos\theta - r\big)\mathrm{d}r \\
&\quad= \left[\frac{r^4}{4}\sin\theta\cos^2\theta - \frac{r^3}{3}\cos^2\theta - \frac{r^3}{3}\cos\theta - \frac{r^3}{3} + \frac{r^2}{2}\cos\theta - \frac{r^2}{2}\right]_0^1 \\
&\quad= \frac{1}{4}\sin\theta\cos^2\theta - \frac{1}{3}\cos^2\theta - \frac{1}{3}\cos\theta - \frac{1}{3} + \frac{1}{2}\cos\theta - \frac{1}{2} \\
&\quad= \frac{1}{4}\sin\theta\cos^2\theta - \frac{1}{3}\cos^2\theta + \frac{1}{6}\cos\theta - \frac{5}{6}.
\end{aligned}$$

Now integrate in θ:

$$\int_0^\pi \sin\theta\cos^2\theta\,\mathrm{d}\theta = \left[-\frac{\cos^3\theta}{3}\right]_0^\pi = \frac{2}{3},$$

$$\int_0^\pi \cos^2\theta\,\mathrm{d}\theta = \frac{\pi}{2}, \qquad \int_0^\pi \cos\theta\,\mathrm{d}\theta = 0, \qquad \int_0^\pi 1\,\mathrm{d}\theta = \pi.$$

Therefore

$$\begin{aligned}
\int_S (\operatorname{curl}\mathbf{F})\cdot\mathbf{N}\,\mathrm{d}\sigma &= \int_0^\pi \left(\frac{1}{4}\sin\theta\cos^2\theta - \frac{1}{3}\cos^2\theta + \frac{1}{6}\cos\theta - \frac{5}{6}\right)\mathrm{d}\theta \\
&= \frac{1}{4}\cdot\frac{2}{3} - \frac{1}{3}\cdot\frac{\pi}{2} + \frac{1}{6}\cdot 0 - \frac{5}{6}\cdot\pi \\
&= \frac{1}{6} - \frac{\pi}{6} - \frac{5\pi}{6} = \frac{1}{6} - \pi.
\end{aligned}$$

Thus

$$\boxed{\int_S (\operatorname{curl}\mathbf{F})\cdot\mathbf{N}\,\mathrm{d}\sigma = \frac{1}{6} - \pi.}$$

(b) Compute $\int_{\partial S} \mathbf{F} \cdot \mathbf{T} \, \mathrm{d}s$

By Stokes' theorem,

$$\int_{\partial S} \mathbf{F} \cdot \mathbf{T} \, \mathrm{d}s = \int_S (\mathrm{curl}\mathbf{F}) \cdot \mathbf{N} \, \mathrm{d}\sigma = \frac{1}{6} - \pi.$$

Hence

$$\boxed{\int_{\partial S} \mathbf{F} \cdot \mathbf{T} \, \mathrm{d}s = \frac{1}{6} - \pi.}$$